Lab 3

Managing Geographies

Introduction to Geopandas & further exploration of the ACS

DATA 20195 Spring 2026

Emma Ross and Michael Xiao

Announcements

- Welcome back to lab!
- **Homework 3 due Wednesday, April 15th, 11:59 pm**, with automatic extension to Friday, April 17th, 1:30pm *(read syllabus for details)*
- Readings
  - **Week 3:** Mark Monmonier, Drawing the Line, Chapter 4 – "Boundary Litigation and the Map as Evidence."
  - **Week 4:** Enrico Moretti, The New Geography of Jobs, Chapter 5 – "The inequality of mobility and cost of living".
  - Printed copies available **up front.**
  - Further optional readings available on Github (see README).
- Office Hours in Ryerson N375B
  - Emma: Tuesdays, 2–3pm
  - Michael: Thursdays, 2–3pm

Warm-Up Question: *what do these numbers represent?*

GEOIDs are the numeric codes the Census uses to name their geographies.

17 031 410900 3 011

State FIPS	County	Tract	Block Group	Block

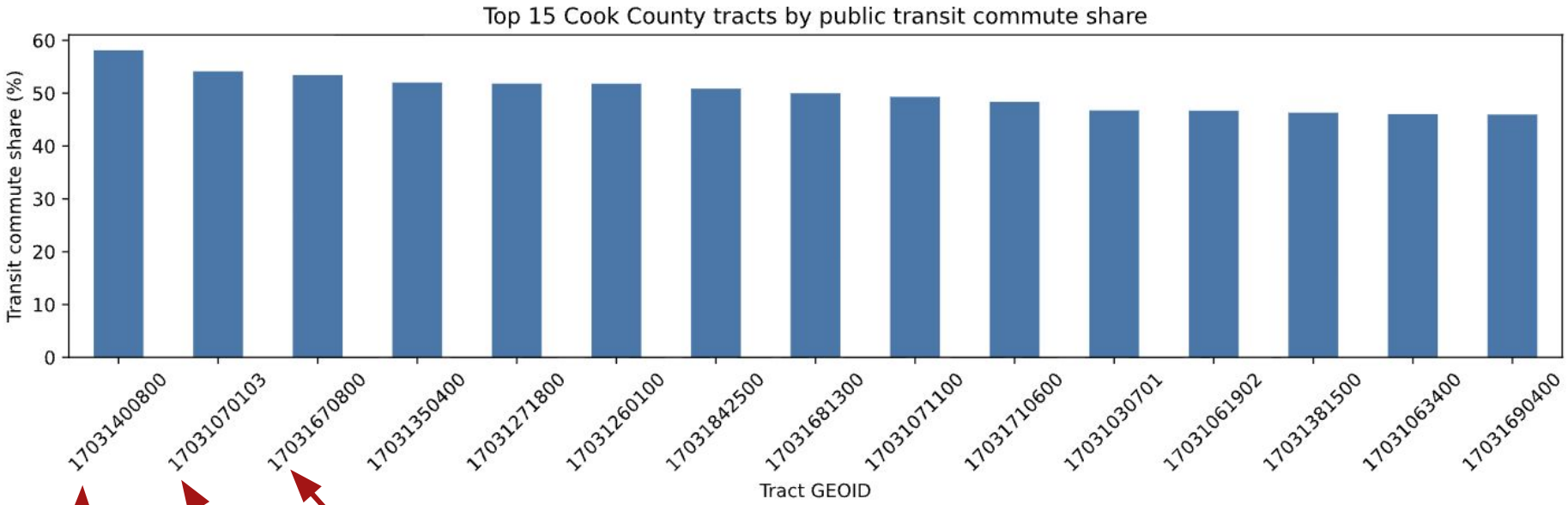
State GEOID: `df.index.str[:2]`

County GEOID: `df.index.str[:5]`

Tract GEOID: `df.index.str[:11]`

Why do we need `.str`?

From last week: how can we improve this plot?



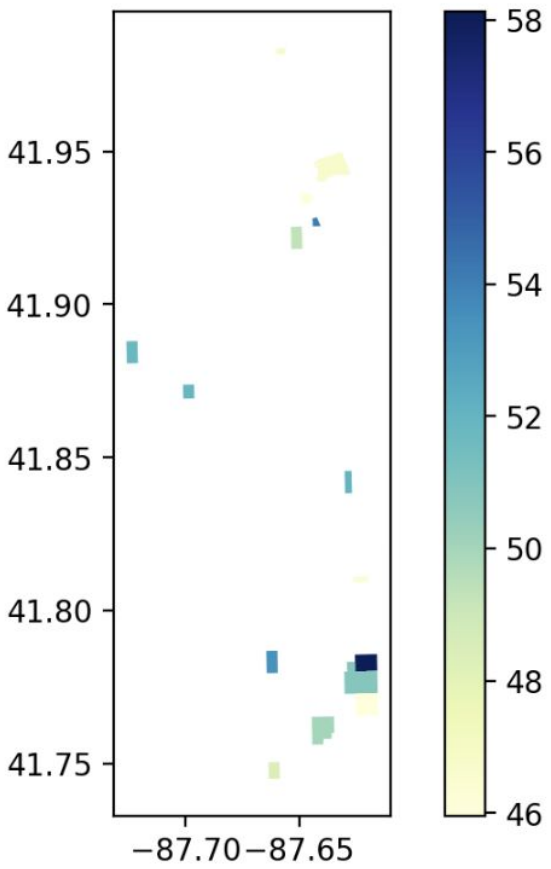
GEOIDs do not easily communicate spatial location.

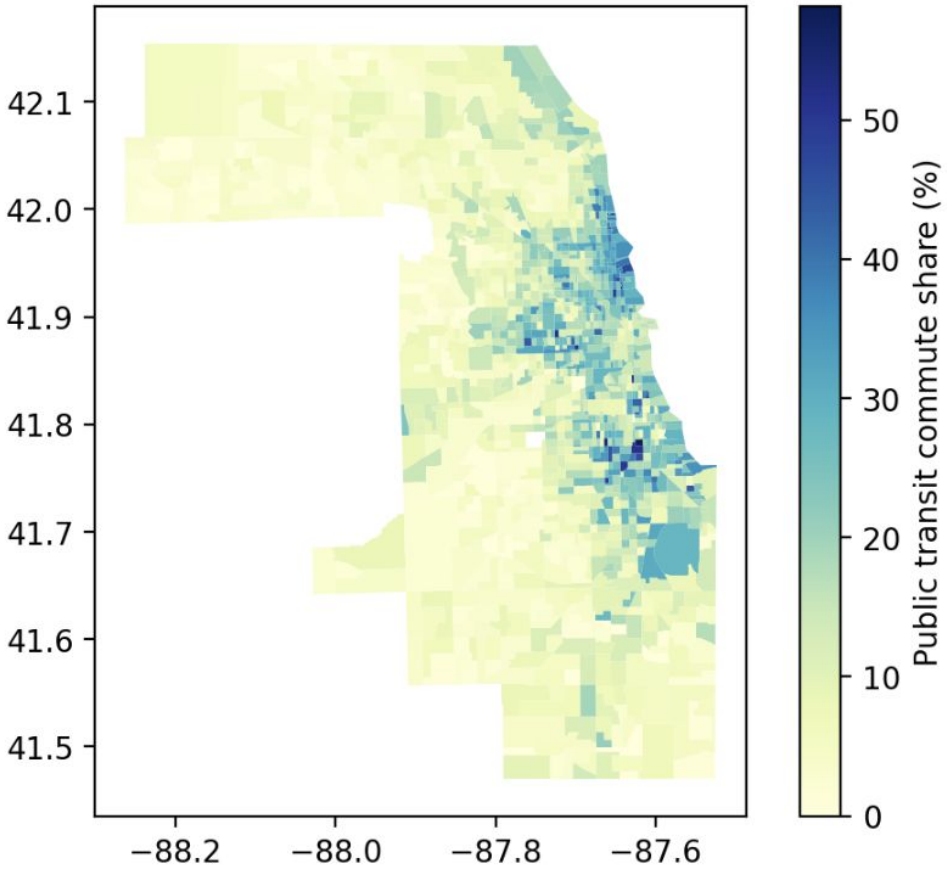
Where is this?

vs. this?

vs. this?

Today we move to visualizations that show the *spatial distribution* of the data.





GeoPandas

`geopandas` (**`gpd`**): Python package that *extends Pandas* to read and store shapefiles

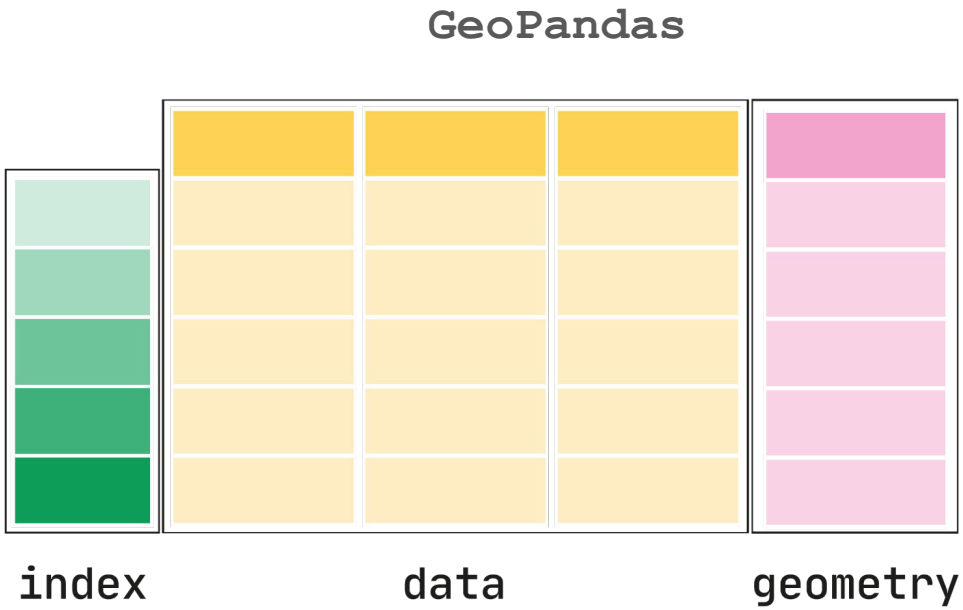
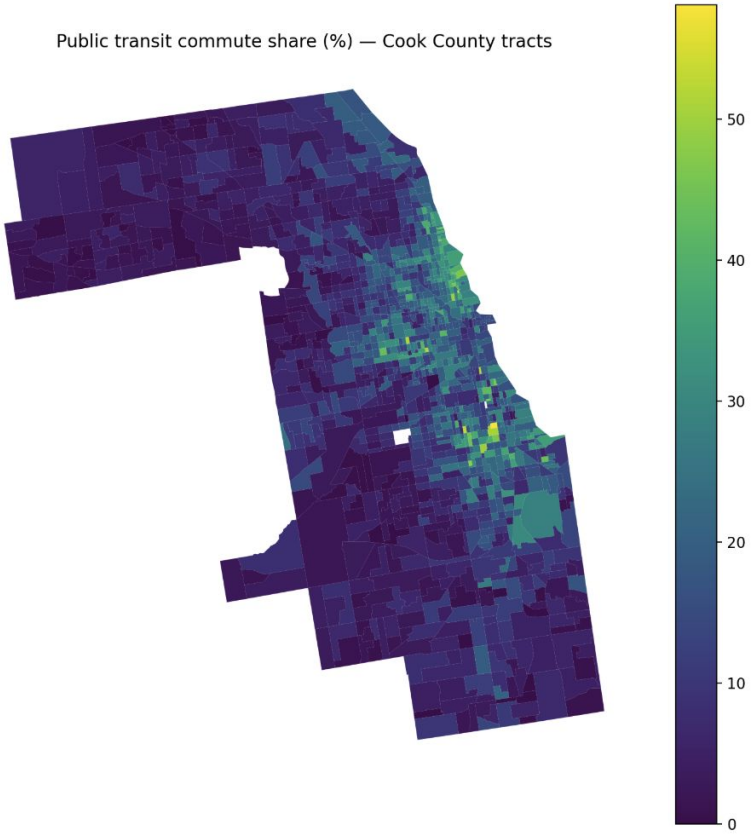


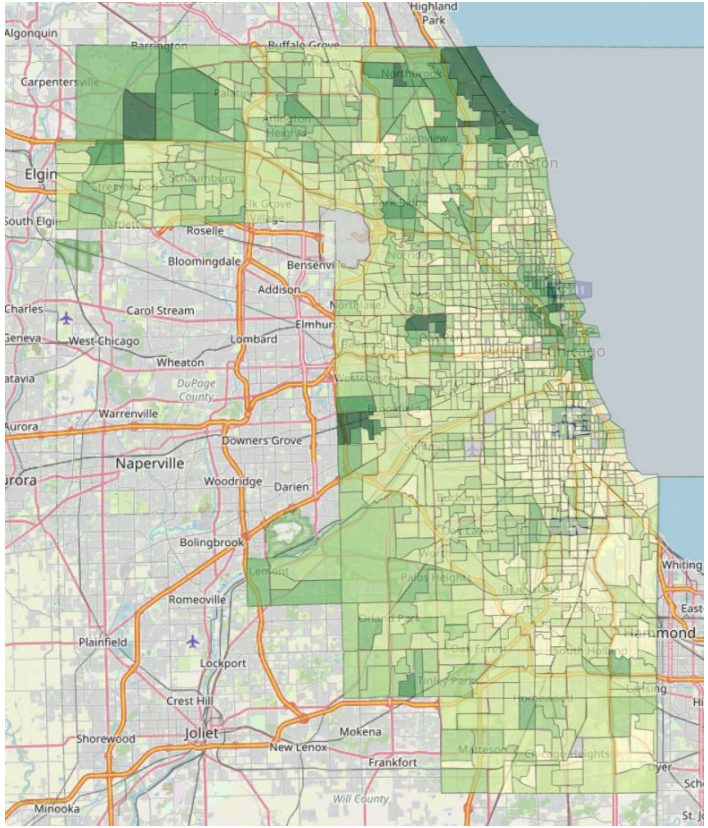
Image Credit: http://geopandas.org/en/stable/getting_started/introduction.html

Maps with GeoPandas

`gdf.plot(...)`



`folium.Choropleth(...)`

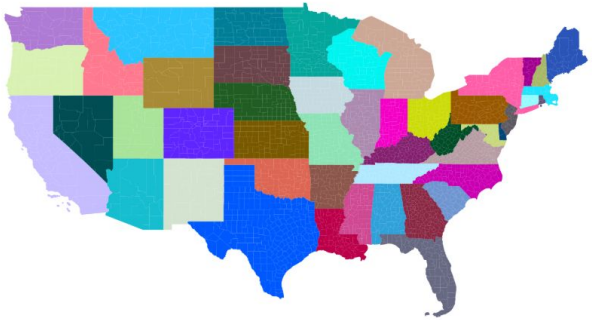


Coordinate Reference Systems & Projections

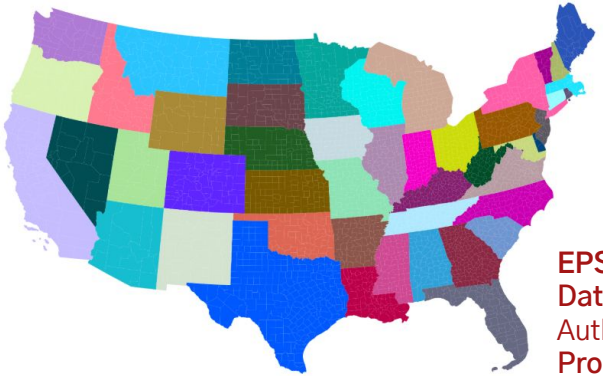
- **Datum:** a mathematical model of the earth as a spheroid
- **Projecting:** act of systematically transposing a **3D** to a **2D** object based on a specific datum

- U.S. Census Bureau uses an un-projected CRS, NAD 83, with center point at the center of mass of Earth and axis tied to the North American tectonic plate

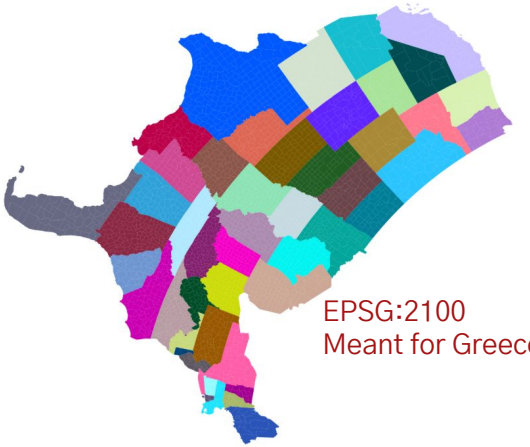
Datum + Projection + Scale = Coordinate Reference System (CRS)



EPSG:4269
Datum: NAD83 (North American Datum of 1983)
Projection: None
*Census default



EPSG:2163
Datum: Not specified (based on Clarke 1866 Authalic Sphere)
Projection: U.S. National Atlas Equal Area



EPSG:2100
Meant for Greece!

(left) `join`

df1

ID		
A	6	ca
B	3	il
C	5	tx

df2

ID			
C	hi	1.1	2
A	hey	3.4	7

`df1.join(df2)`

A	6	ca	hey	3.4	7
B	3	il	*nan*	*nan*	*nan*
C	5	tx	hi	1.1	2

Demo adapted from UC Berkeley's Data 100 (Principles and Techniques of Data Science) Summer 2025

(left) `join`

df1

ID		
A	6	ca
B	3	il
C	5	tx

df2

ID			
C	hi	1.1	2
A	hey	3.4	7
B	yay	2.5	9
B	wow	7.3	1

`df1.join(df2)`

A	6	ca	hey	3.4	7
B	3	il	yay	2.5	9
B	3	il	wow	7.3	1
C	5	tx	hi	1.1	2

Demo adapted from UC Berkeley's Data 100 (Principles and Techniques of Data Science) Summer 2025

apply

df =

	Col1	Col2	Col3
	cat	3	1
	dog	1	1
	owl	4	3
	fly	1	1
	uno	5	2
	pad	9	2
	pen	2	3
	key	5	2
	see	6	1

```
def larger3(x):
    return max(x,3)
```

```
df['Col2'].apply(larger3)
```

→

3
3
4
3
5
9
3
5
6

apply

df =

Col1	Col2	Col3
cat	3	1
dog	1	1
owl	4	3
fly	1	1
uno	5	2
pad	9	2
pen	2	3
key	5	2
see	6	1

```
def rpl_str(x):
    if x[1] == 'a':
        return 'cat'
    elif x[2] == 'y':
        return 'sty'
    else:
        return x
```

```
df['Col1'].apply(rpl_str)
```

→

cat
dog
owl
sty
uno
cat
pen
sty
see

pandas operation: `.unique()`

df =

Col1	Col2	Col3
cat	3	1
dog	1	1
cat	4	3
fly	1	1
frog	5	2
cat	9	2
dog	2	3
owl	5	2
fly	6	1

`df['Col1'].unique()` →

cat
dog
fly
frog
owl

Why is ACS data reported at the *tract* level instead of *county subdivisions* or *blocks*?

Boston Election

Precincts

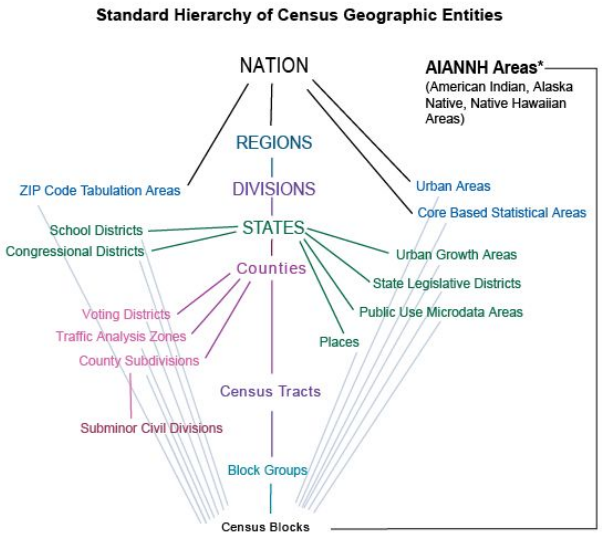
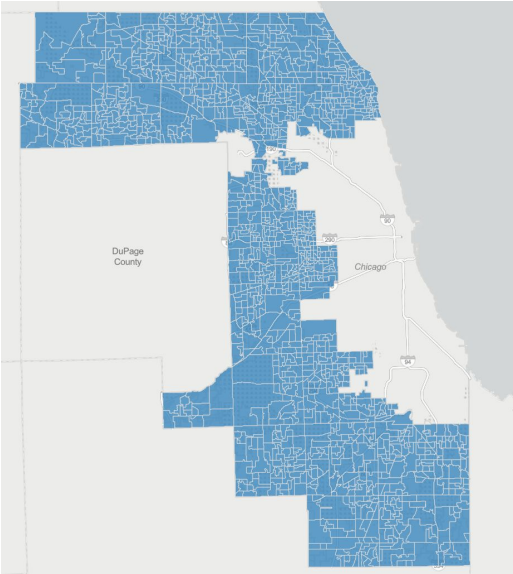


Image credit: University of Missouri

Who can find precincts on the central spine?

Election precinct boundaries are **drawn by local officials for administrative purposes** — they freely cross tract and block group lines – and sometimes even blocks!

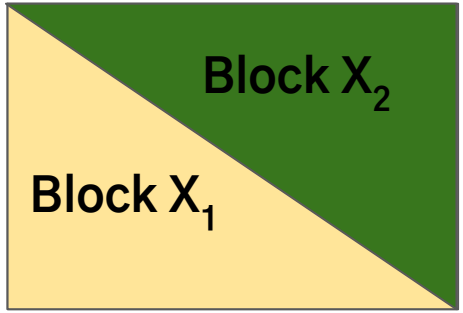


Proration

Citizen Voting-Age Population (CVAP)

700

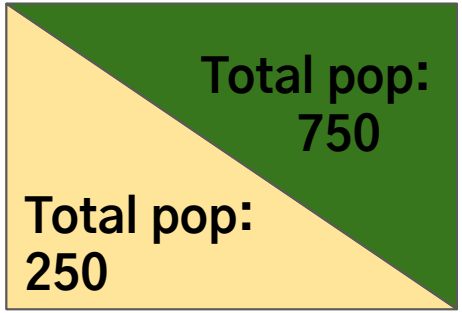
Proration by area



Block X_1: 350

Block X_2: 350

Proration by population density



Block X_1: 175

Block X_2: 525

Election Data Misalignment

- What oddities or concerns do you have about the Boston election data?
- What difference do you observe between the election data and the shapefile?
- How can we construct a “precinct index”?

	City/Town	Ward	Pct	Obama/ Biden	Romney/ Ryan	Johnson/ Gray	Stein/ Honkala	All Others	Blanks	Total Votes Cast
0	NaN	NaN	NaN	Democratic	Republican	Libertarian	Green-Rainbow	NaN	NaN	NaN
1	Abington	-	1	844	705	6	7	4	5	1,571
2	Abington	-	2	742	741	18	8	4	7	1,520
3	Abington	-	3	815	801	13	11	2	6	1,648
4	Abington	-	4	899	904	8	9	5	5	1,830
...	...	...	...	...	...	...	...	...	...	...
2171	Yarmouth	-	4	1,056	909	10	9	2	15	2,001
2172	Yarmouth	-	5	882	810	11	8	0	8	1,719
2173	Yarmouth	-	6	924	826	13	11	2	7	1,783
2174	Yarmouth	-	7	1,169	1,269	13	11	1	6	2,469
2175	TOTALS	NaN	NaN	1,921,290	1,188,314	30,920	20,691	6,552	16,429	3,184,196

CVAP Proration

Our calculation:

$$\mathrm{CVAP}(block\ X) = \frac{\mathrm{VAP}_{PL}(block\ X)}{\mathrm{VAP}_{PL}(tract\ Y)} \cdot \mathrm{CVAP}_{ACS}(tract\ Y)$$

DDRI calculation:

$$\mathrm{CVAP}(block\ X) = \frac{\mathrm{CVAP}_{ACS}(tract\ Y)}{\mathrm{VAP}_{ACS}(tract\ Y)} \cdot \mathrm{VAP}_{PL}(block\ X)$$

with the exception of when $\mathrm{VAP}_{ACS}(tract\ Y) < 20$, in which case

$$\mathrm{CVAP}(block\ X) = \frac{\mathrm{CVAP}_{ACS}(state\ Z)}{\mathrm{VAP}_{ACS}(state\ Z)} \cdot \mathrm{VAP}_{PL}(block\ X)$$

Data Cleaning

What did we do to process the Boston election dataset?

- Remove extra rows (first and last)
- Convert vote count to `int` (remove commas)
- Reformat missing values ("–" to `None`)
- Match and modify invalid/malformed precinct names
- …

Data cleaning/processing takes up a **large** (if not majority) portion of the time for data projects.

Every **decision** we make can potentially impact the analysis result. **Recording** these judgement calls is crucial.